

Giovanni Rosa da Silva

Software Technical Lead | IoT & Distributed Systems Specialist

Brazilian | Porto Belo, SC – Brazil

✉ giovanni_rosa4@hotmail.com | ☎ +55 41 99610-1950

🌐 <https://gio-dev.vercel.app> | GitHub: <https://github.com/giovannirosa>

Professional Summary

Software Technical Lead specializing in IoT platforms, distributed systems, and scalable cloud-based architectures. Proven track record leading international engineering teams, defining system architecture, and delivering production-grade solutions across aviation, healthcare, finance, real estate, and industrial IoT. Extensive experience integrating cloud-native services, real-time data pipelines, and device communication protocols in high-availability systems.

Professional Experience

Software Technical Lead

GRDS IT Services (Technology Consulting) | Dec 2022 – Present

- Lead a software engineering team delivering international IoT asset monitoring and control platforms used in production environments.
 - Define system architecture and technical direction for distributed, cloud-based applications handling real-time device data.
 - Design and oversee solutions using Next.js, Node.js, MQTT, InfluxDB, and REST APIs.
 - Integrate cloud-native services across **AWS and Azure**, including IoT Core, serverless compute, data streaming, managed databases, and secure API infrastructure.
 - Manage developers using Agile/Scrum methodologies, ensuring high standards of code quality, scalability, reliability, and roadmap execution.
-

Senior Software Engineer

CI&T | Mar 2015 – Dec 2022

- Designed and developed software solutions for international and domestic projects across aviation, healthcare, IoT, finance, and real estate sectors.
- Collaborated directly with global clients such as **SITA, Schneider, Corvalent, and RCI Bank**, contributing to system design, implementation, and delivery of production-grade applications.
- Played a key role in modernizing legacy systems, including migrations to modern frontend frameworks and backend architectures.

- Delivered scalable web and backend solutions using Java, Spring Boot, Node.js, Python, and modern JavaScript frameworks.
 - Contributed to technical decision-making, performance optimization, and cross-team collaboration in distributed environments.
-

Education

Master of Science in Networks and Distributed Systems

Federal University of Paraná (UFPR) | 2020 – 2023

Bachelor of Science in Computer Science

Federal University of Paraná (UFPR) | 2014 – 2019

Publications & Academic Contributions

- [*Adaptive Access Control for Smart Homes Supported by Zero Trust for User Actions*](#), IEEE Transactions on Network and Service Management (TNSM), 2024.
 - [*Context-aware and User Behavior-based Continuous Authentication for Zero Trust Access Control in Smart Homes*](#), UFPR, 2023.
 - [*Zero Trust Access Control with Context-Aware and Behavior-Based Continuous Authentication for Smart Homes*](#), SBSeg, 2021.
 - [*Coordenação de Múltiplos Eventos Críticos de Saúde por Redes Dinâmicas de Interesses Sociais*](#), SBRC, 2020.
 - [*Coordenação de Eventos Críticos Concorrentes para Suportar Tomadas de Decisão na Disseminação de Dados Pessoais Sensíveis*](#), UFPR, 2020
-

Technical Expertise

Core Specializations

- Distributed Systems & IoT Platforms
- Cloud-Native Architectures & High-Availability Systems
- Software Architecture & System Design
- Technical Leadership & Agile/Scrum Practices

Technologies

- **Languages & Frameworks:** Java, JavaScript, Python, Node.js, React, Next.js, AngularJS, Angular, Vue.js, React Native
- **Backend & APIs:** Spring Boot, REST APIs, MQTT, Django REST Framework
- **Databases:** PostgreSQL, MySQL, InfluxDB, MongoDB, DynamoDB
- **Cloud & DevOps:** AWS, Azure, Google Cloud, Docker, CI/CD pipelines
- **Tools:** Git, Postman, Figma

Certifications

- [Oracle Certified Professional, Java SE 8 Programmer II](#) (2017)
 - [Oracle Certified Associate, Java SE 8 Programmer I](#) (2016)
-

Languages

- **Portuguese:** Native
 - **English:** Proficient (TOEIC 975/990 | TOEFL 580/677)
 - **Spanish:** Intermediate (B1)
 - **French:** Beginner
 - **Italian:** Beginner
-

Selected Projects

- Designed and delivered IoT management platforms integrating real-time device data, cloud services, and scalable APIs.
 - Led modernization initiatives, migrating legacy enterprise systems to modern, maintainable architectures.
 - Developed internal tooling and automation systems supporting data processing and operational workflows.
-

Professional Focus

IoT systems, distributed architectures, IoT security, machine learning integration in production systems, and scalable cloud-native platforms.